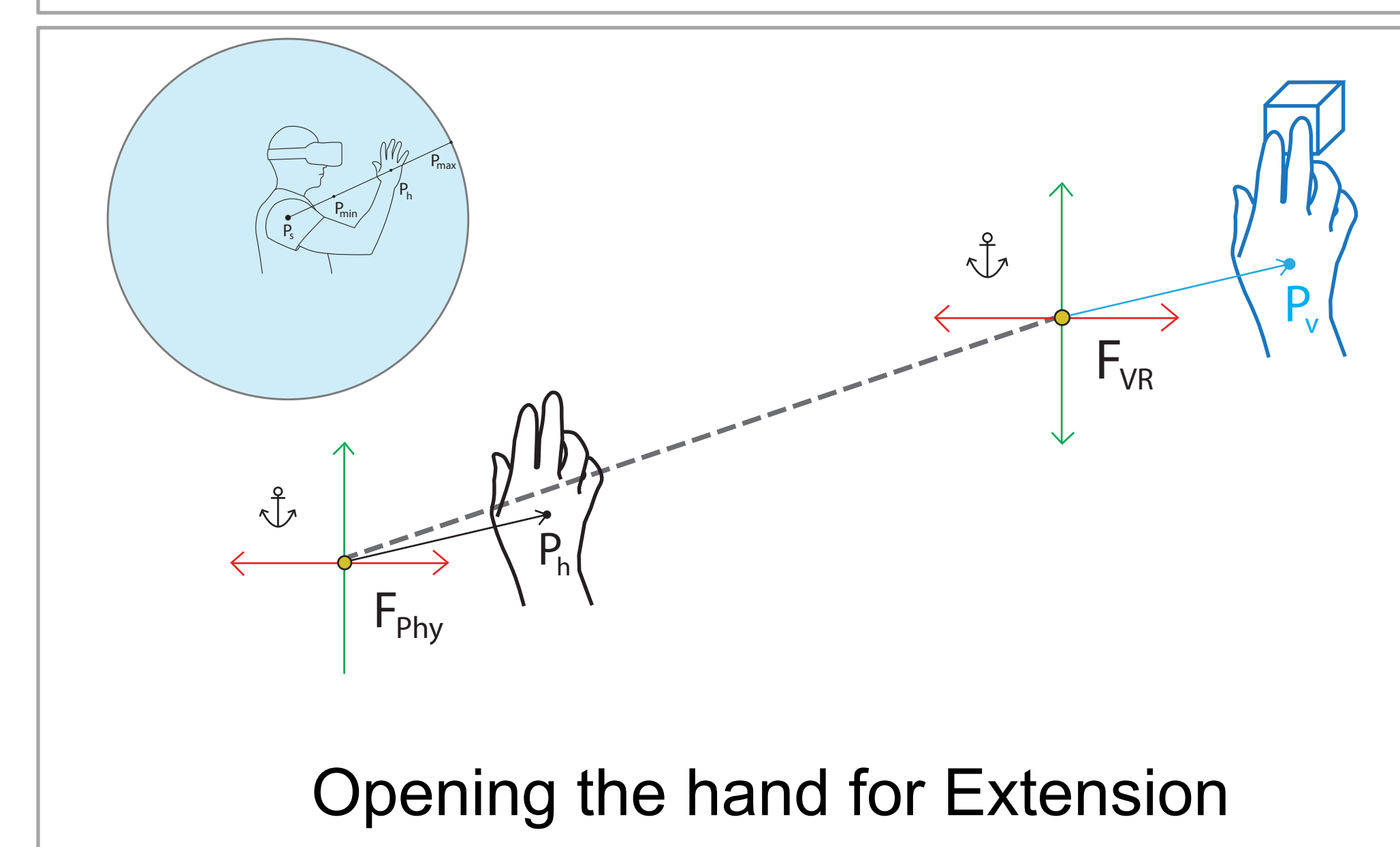
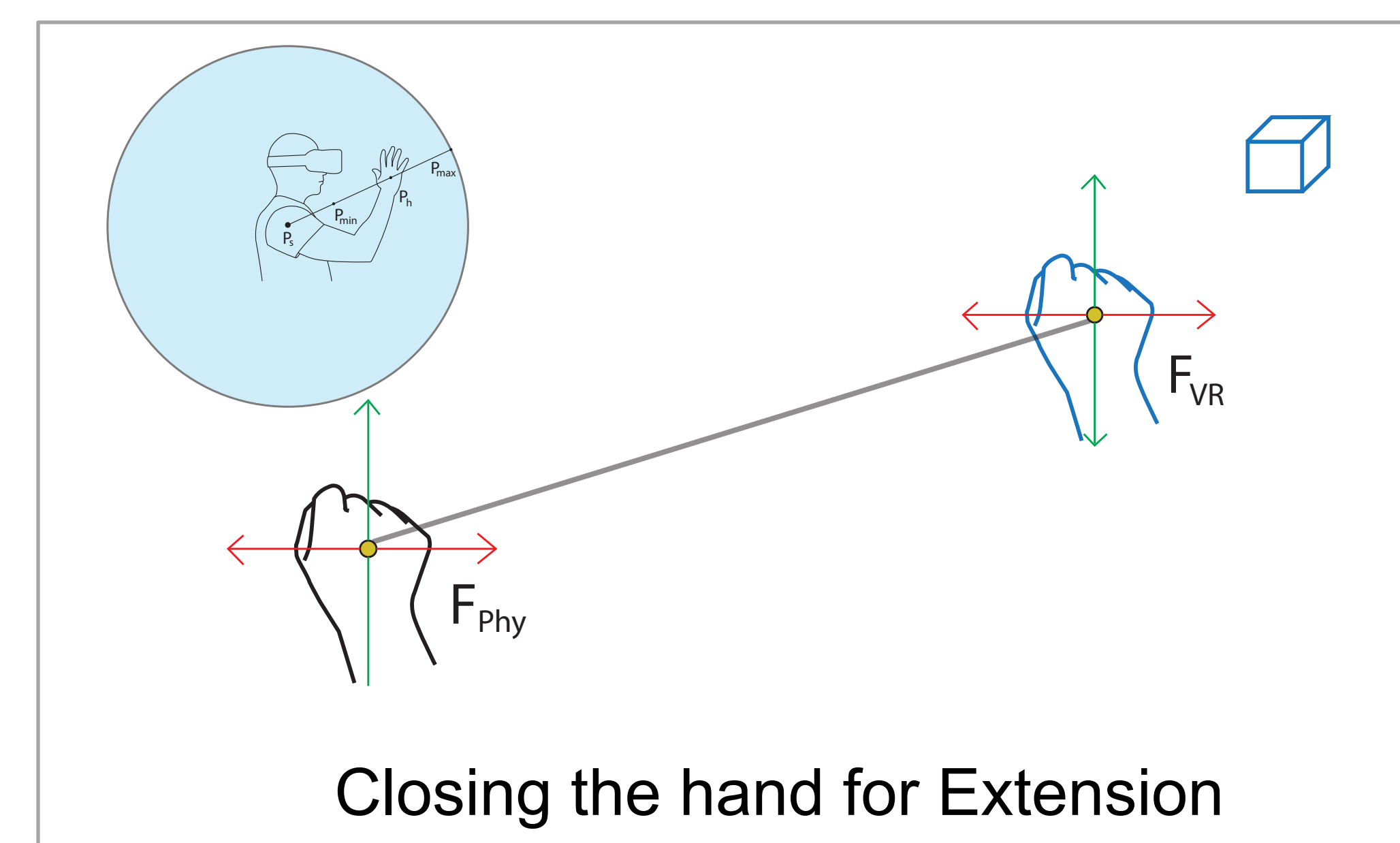
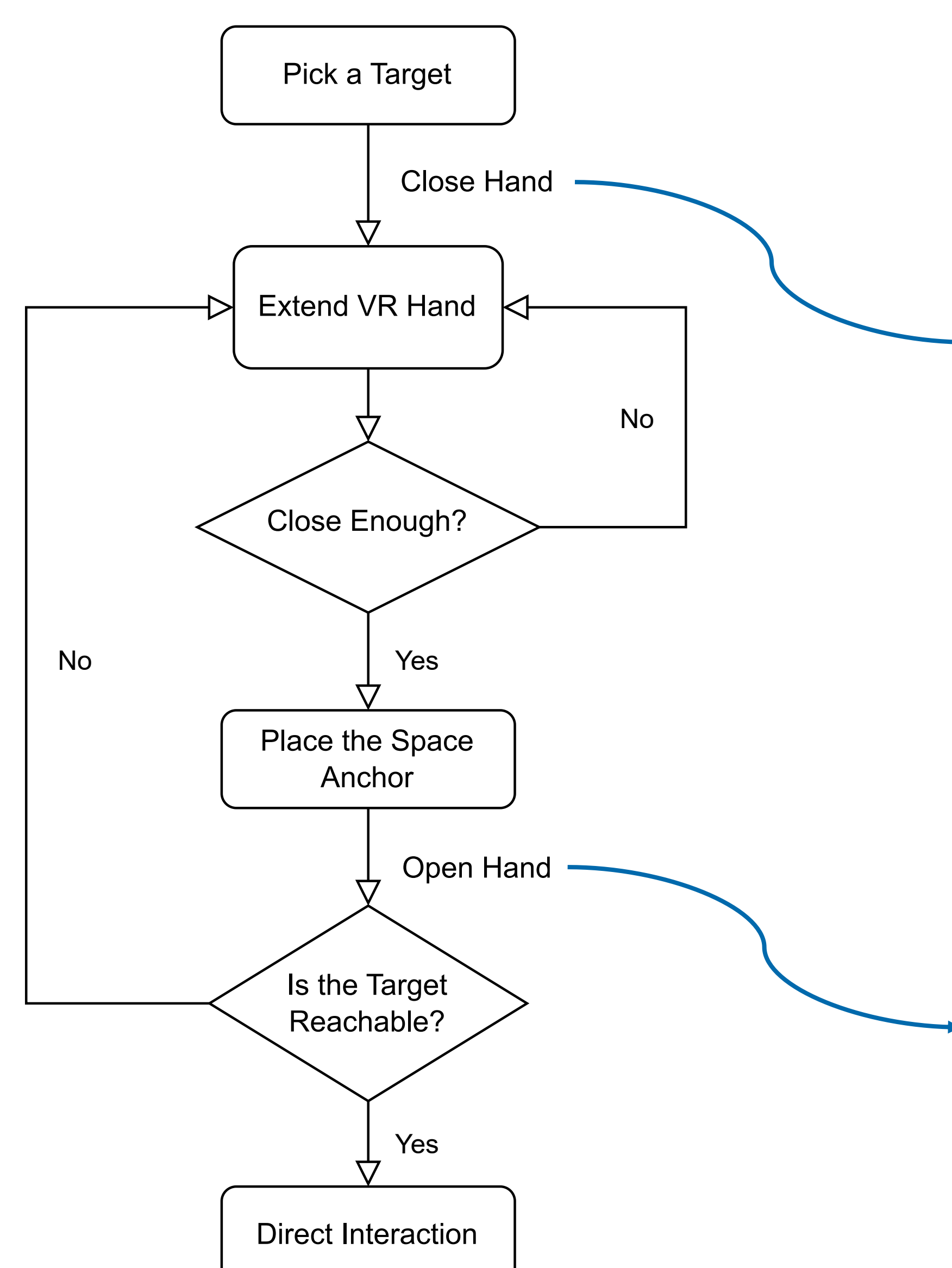


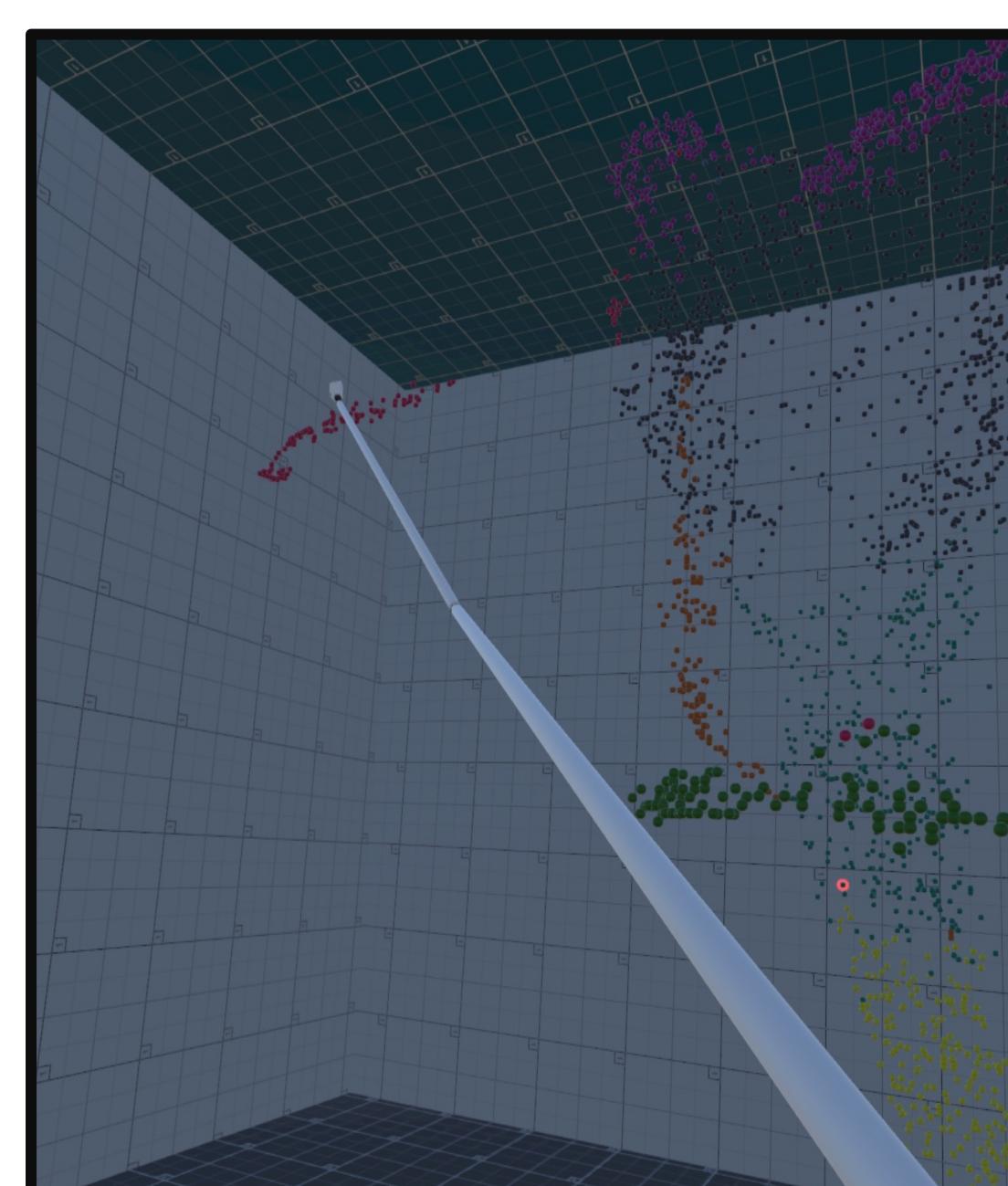


Introduction

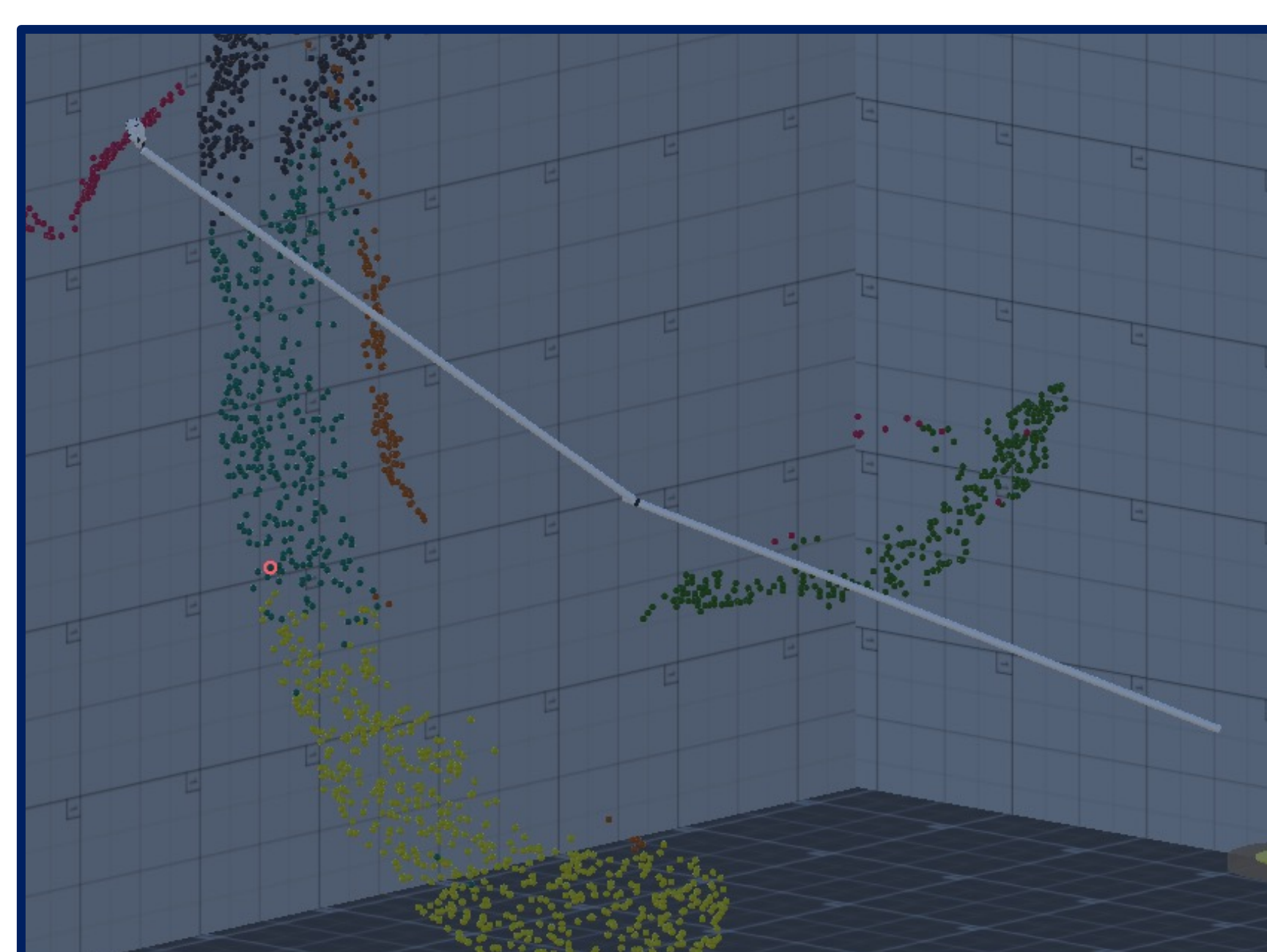
In the immersive world of Virtual Reality (VR), where endless virtual spaces beckon, we present a solution that unites you with far-off elements. Conventional VR interaction methods have grappled with accuracy and ease. Techniques like ray casting and teleportation had their moments [1,2], but fell short on both fronts. Enter our innovation: extending your virtual hand to touch the unreachable. By placing a space anchor near the target, we harmonize your physical and virtual hand movements. The result? A fusion of natural intuition and precise control. Bid farewell to compromises. Our novel approach redefines VR interaction, making distant elements accessible with a virtual reach. Join us in embracing a new era of exploration.



Extended Hand



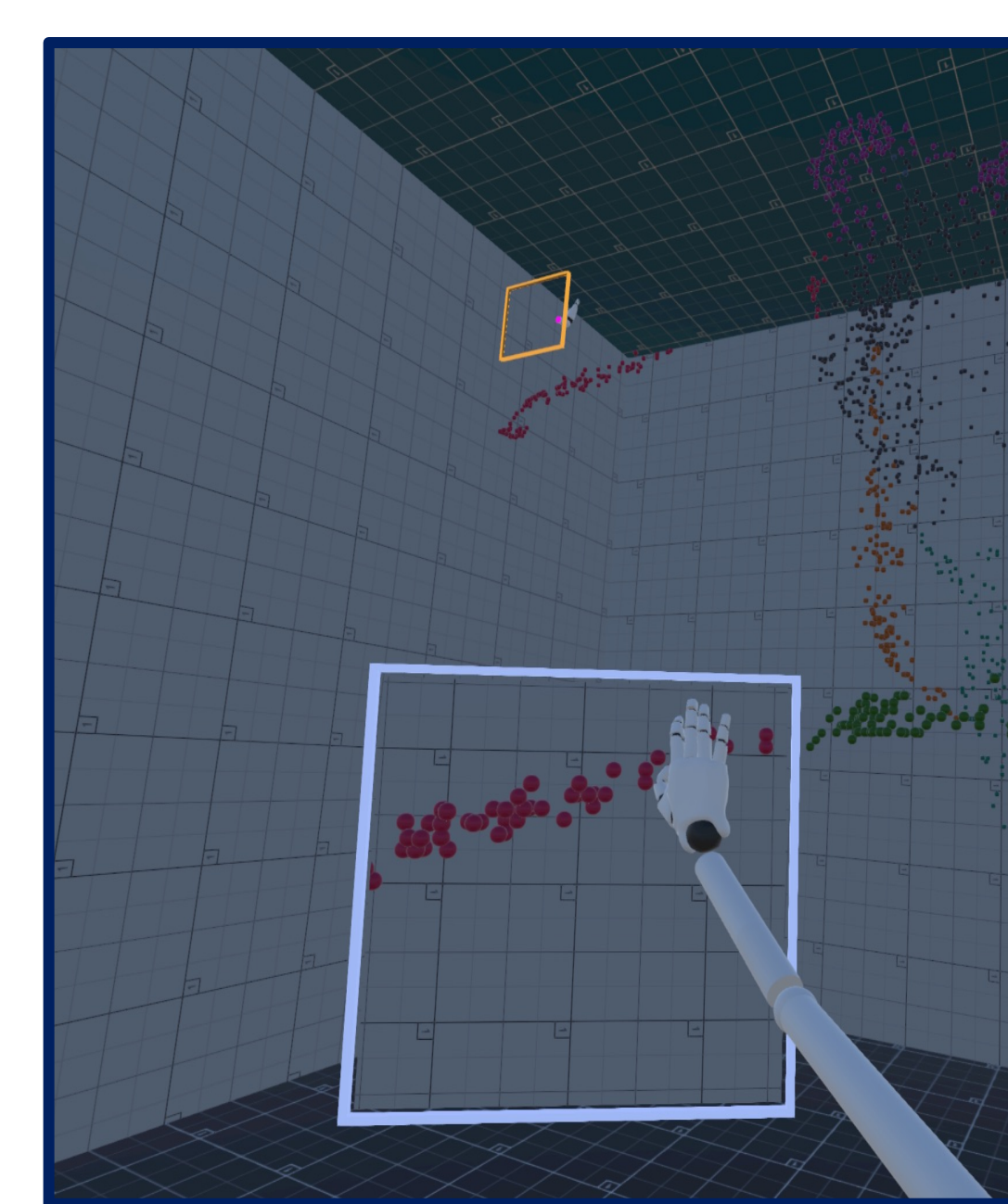
VR Headset View



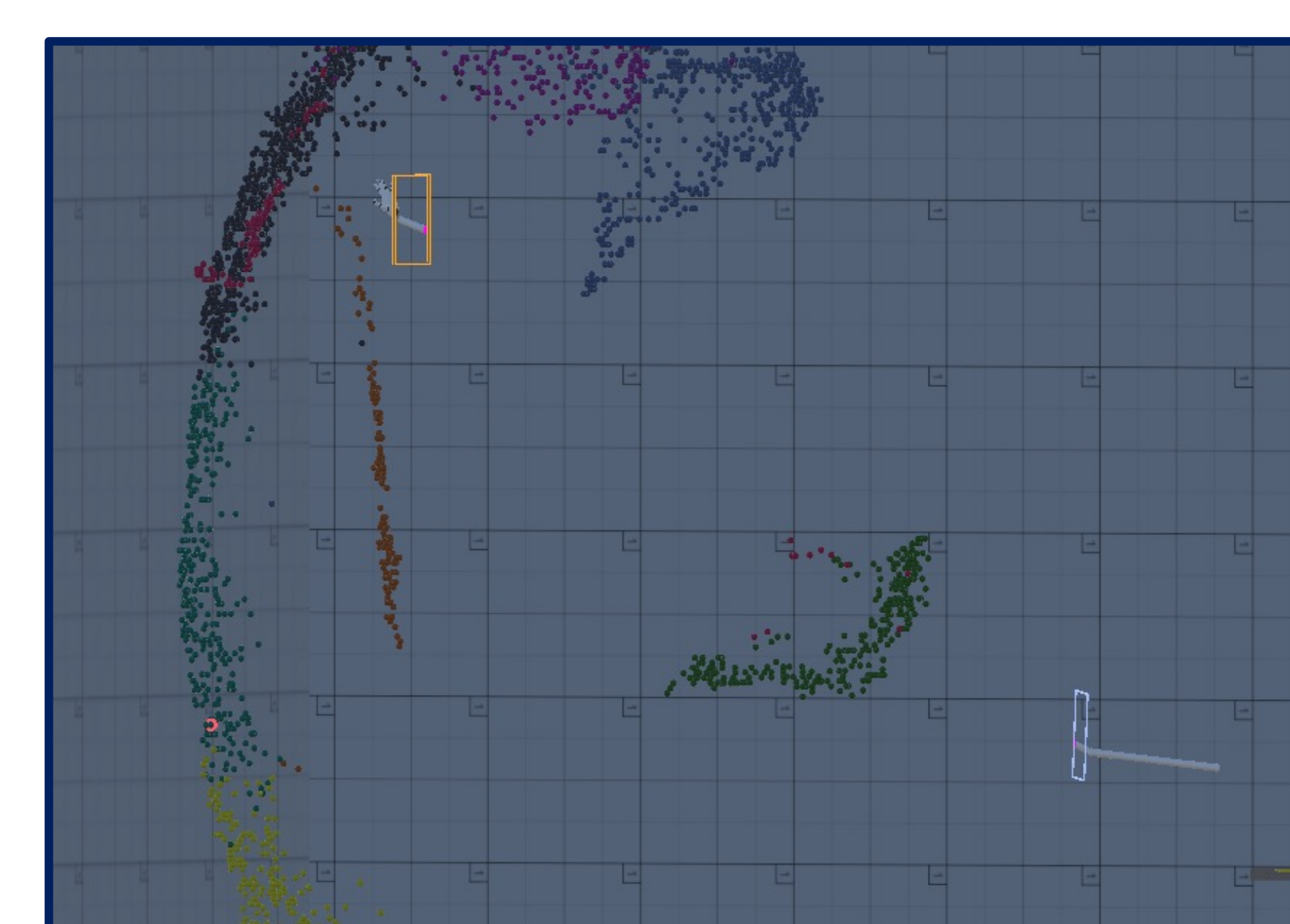
VR Side View

In the extended-hand scenario, while interacting with the virtual object, the user's hand remains extended, and the space anchor is rendered invisible. Users are required to gauge the depth perspective between their virtual hand and the target during the interaction without any aid.

Portals



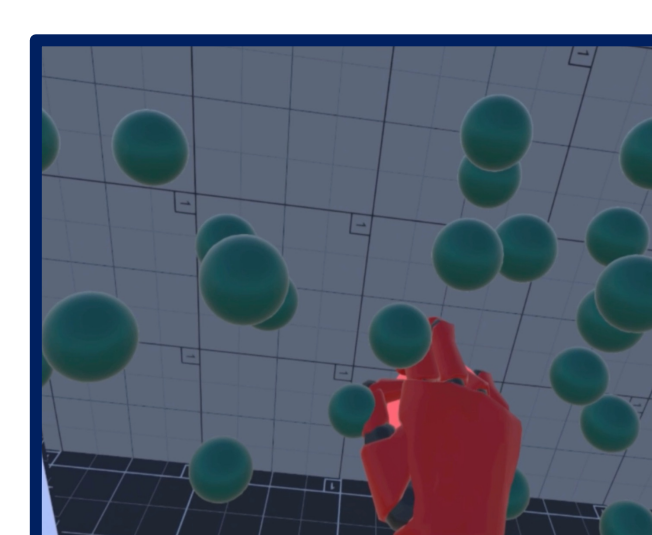
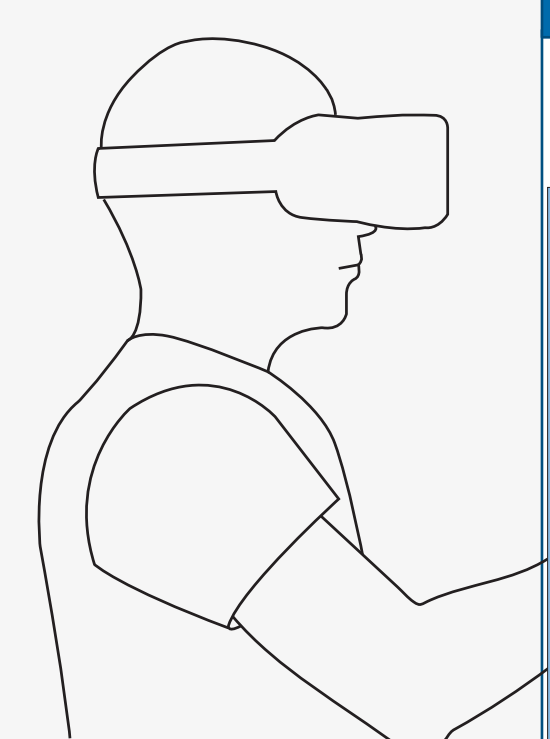
VR Headset View



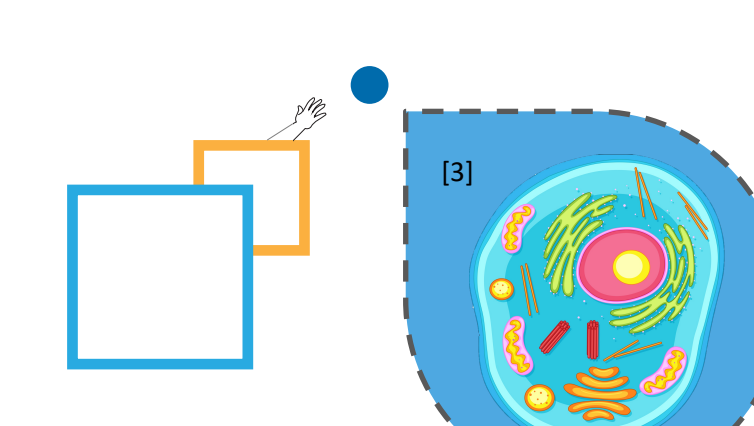
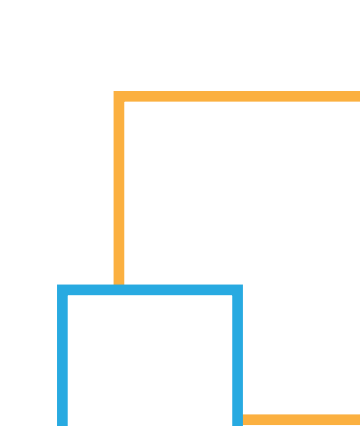
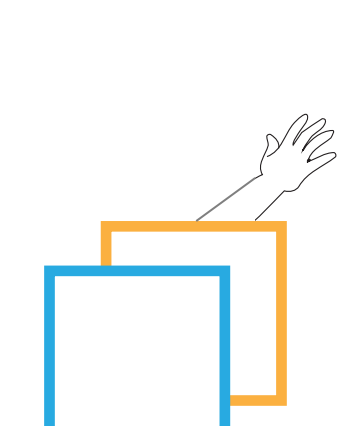
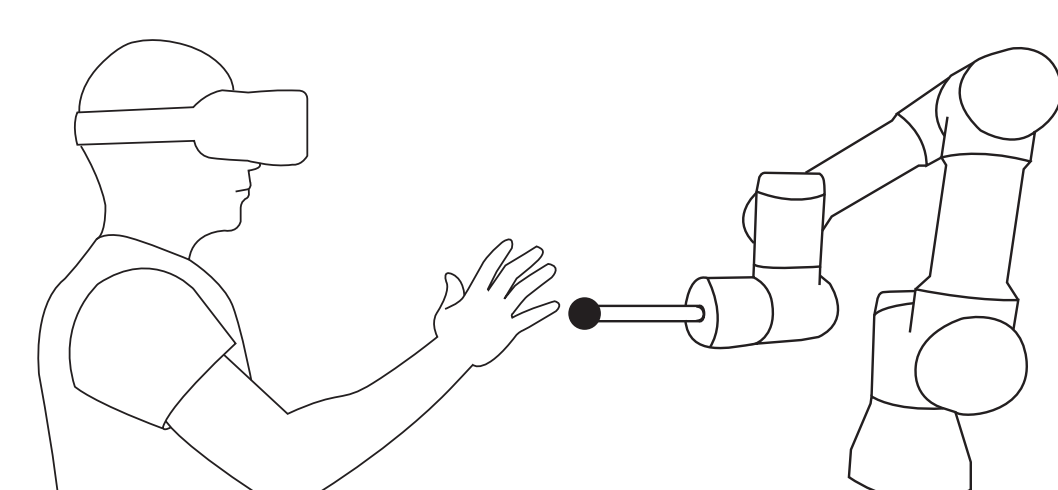
VR Side View

In the portal scenario, a pair of portals represents the space anchor, and user's virtual hand is scaled back to normal during the interaction. Users navigate the spatial divide by putting their hand through the portal, effectively bridging the gap to the target.

Tangible Feedback and Other Features for Interactions Through Portal



Tangible Feedback



Functional Portal Size

Contact Information

Shaozhang Dai
Monash University
Email: Shaozhang.dai1@monash.edu

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2. I. Poupyrev, M. Billingham, S. Weghorst, and T. Ichikawa. The go-go interaction technique: non-linear mapping for direct manipulation in vr. In *Proceedings of the 9th annual ACM symposium on User interface software and technology*, pp. 79–80, 1996.
3. Image by brgfx / Freepik

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